Let $B = (0,0)$ and $C = (17,0)$. From $AB = 26$ and $AC = 25$, we may take

$$A = (10, 24).$$

Since $26^2 < 25^2 + 17^2$, the triangle is acute.

0.0.1 1. Finding X

By the cosine rule,

$$\cos B = \frac{26^2 + 17^2 - 25^2}{2 \cdot 26 \cdot 17} = \frac{85}{221}.$$

Therefore

$$BF = BC \cos B = \frac{85}{13}, \qquad AF = 26 - \frac{85}{13} = \frac{253}{13}.$$

Similarly,

$$\cos C = \frac{25^2 + 17^2 - 26^2}{2 \cdot 25 \cdot 17} = \frac{119}{425},$$

so

$$CE = BC \cos C = \frac{119}{25}, \qquad AE = 25 - \frac{119}{25} = \frac{506}{25}.$$

Applying Menelaus' theorem to triangle ABC and the collinear points F, E, X,

$$\frac{AF}{FB} \cdot \frac{BX}{XC} \cdot \frac{CE}{EA} = 1.$$

Hence

$$\frac{BX}{XC} = \frac{FB}{AF} \cdot \frac{EA}{CE} = \frac{85}{253} \cdot \frac{506}{119} = \frac{10}{7}.$$

Thus X lies beyond C, and since $BX - XC = BC = 17$,

$$BX = \frac{170}{3}, \qquad XC = \frac{119}{3}.$$

Therefore

$$X = \left(\frac{170}{3}, 0 \right).$$

0.0.2 2. Finding AQ and HQ

We have

$$\overrightarrow{AX} = \left(\frac{140}{3}, -24 \right) = \frac{4}{3}(35, -18),$$

so

$$AX = \frac{4}{3}\sqrt{35^2 + 18^2} = \frac{4\sqrt{1549}}{3}.$$

Using the power of X with respect to Ω,

$$XA \cdot XQ = XB \cdot XC.$$

Thus

$$XQ = \frac{\frac{170}{3} \cdot \frac{119}{3}}{\frac{4\sqrt{1549}}{3}} = \frac{10115}{6\sqrt{1549}}.$$

This is less than XA, so Q lies between X and A. Hence

$$AQ = AX - XQ = \frac{4\sqrt{1549}}{3} - \frac{10115}{6\sqrt{1549}} = \frac{759}{2\sqrt{1549}}.$$

Since AQ has direction $(35, -18)$,

$$\overrightarrow{AQ} = \frac{759}{2 \cdot 1549}(35, -18).$$

The altitude from A is $x = 10$. Since AC has direction $(7, -24)$, the altitude from B has slope $7/24$. Therefore

$$H = \left(10, \frac{35}{12}\right),$$

and

$$\overrightarrow{AH} = \left(0, -\frac{253}{12}\right).$$

Using $759 = 3 \cdot 253$,

$$\overrightarrow{HQ} = \overrightarrow{AQ} - \overrightarrow{AH} \quad = \frac{3 \cdot 253}{2 \cdot 1549}(35, -18) + \left(0, \frac{253}{12}\right) \quad = \frac{253 \cdot 35}{12 \cdot 1549}(18, 35).$$

Because $18^2 + 35^2 = 1549$,

$$HQ = \frac{253 \cdot 35}{12\sqrt{1549}}.$$

0.0.3 3. Interpreting P using the nine-point circle

Let T be the reflection of H across AC. The reflection of the orthocentre across any side lies on the circumcircle; indeed,

$$\angle ATC = \angle AHC = 180^\circ - \angle ABC.$$

Thus $T \in \Omega$. Also, since E is the perpendicular foot on AC, E is the midpoint of HT.

Consider the homothety centred at H with ratio $1/2$. The reflections of H across BC, CA, AB lie on Ω, and their images under this homothety are respectively D, E, F. Hence the homothety sends Ω to circle DEF.

Therefore the image of Q is the midpoint of HQ, which lies on circle DEF. Since the triangle is acute, H lies inside this circle, so this midpoint is the unique intersection of the ray HQ with circle DEF. Consequently,

$$P \text{ is the midpoint of } HQ.$$

The same homothety sends T to E. Therefore

$$PQ = \frac{HQ}{2}, \qquad PE = \frac{QT}{2},$$

and hence

$$\frac{PQ}{PE} = \frac{HQ}{QT}.$$

0.0.4 4. Finding QT

A unit vector along AC is

$$\mathbf{u} = \frac{1}{25}(7, -24).$$

The direction from A to H is $\mathbf{d} = (0, -1)$. Reflecting $\mathbf{d}$ across the line in direction $\mathbf{u}$ gives

$$\mathbf{d}' = 2(\mathbf{d} \cdot \mathbf{u})\mathbf{u} - \mathbf{d} \quad = 2\left(\frac{24}{25}\right)\frac{1}{25}(7, -24) - (0, -1) \; = \frac{1}{625}(336, -527).$$

Thus AT has direction $(336, -527)$, while AQ has direction $(35, -18)$. Therefore

$$\sin \angle QAT = \frac{|35(-527) - (-18)(336)|}{\sqrt{1549} \cdot 625} \quad = \frac{12397}{625\sqrt{1549}} = \frac{49 \cdot 253}{625\sqrt{1549}}.$$

The area of ABC is

$$[ABC] = \frac{1}{2} \cdot 17 \cdot 24 = 204,$$

so its circumradius is

$$R = \frac{26 \cdot 25 \cdot 17}{4 \cdot 204} = \frac{325}{24}.$$

Since A, Q, T lie on Ω, the extended sine rule gives

$$QT = 2R \sin \angle QAT \quad = \frac{325}{12} \cdot \frac{49 \cdot 253}{625\sqrt{1549}} \; = \frac{253 \cdot 637}{300\sqrt{1549}}.$$

Combining 1, 2, and 3,

$$\frac{PQ}{PE} = \frac{\frac{253 \cdot 35}{12\sqrt{1549}}}{\frac{253 \cdot 637}{300\sqrt{1549}}} = \frac{35 \cdot 25}{637} = \frac{125}{91}.$$

Thus $a = 125$, $b = 91$, and

$$a + b = 216.$$